Mathematical and Computational Methods

Unit 10: Integration II: techniques and applications

Unit 9 built the definite integral, the Fundamental Theorem, and the workhorse method of substitution. This unit completes the toolkit and then puts it to use. We add the three techniques that reach the integrands substitution alone cannot—*integration by parts* (the product rule reversed), *partial fractions* (for rational functions), and the *trigonometric methods* (identities, and trigonometric substitution)—and then turn integration loose on the problems it was made for: *areas between curves*, *volumes of revolution*, *average values*, and the *expectation* of a probability distribution. The last of these carries us, finally, to integrals that run to infinity.

Where this fits. Each technique mirrors a rule of differentiation: by parts reverses the product rule of Unit 7, just as substitution reversed the chain rule. Partial fractions integrate the rational functions whose graphs you sketched in Unit 8, and the trigonometric methods lean on the identities of Units 1 and 3 and the inverse-trigonometric integrals tabulated in Unit 9. The applications build directly on the area reading of the definite integral (Unit 9) and the curve analysis of Unit 8; the probability strand ties together the data-science thread of the course (the sigmoid of Unit 1, gradient descent in Unit 8, the Gaussian of Unit 9) and the physics one (decay, work). This is the last calculus unit, and the last of the module; Taylor series and differential equations remain with the dedicated Calculus module.

Contents

Learning objectives

By the end of this unit you will be able to:

- Integrate by parts, by partial fractions, and by trigonometric methods.
- Compute areas between curves and volumes of revolution.
- Find the average value of a function.
- Evaluate improper integrals and find probabilities and expectations from a density.
- Set up definite integrals from a geometric or physical description.

1 Further techniques

1.1 Integration by parts

Substitution reverses the chain rule. The remaining workhorse, *integration by parts*, reverses the *product* rule. Starting from $(uv)' = u'v + uv'$ of Unit 7 and integrating both sides, $uv = \int u'v\,\mathrm{d}x + \int uv'\,\mathrm{d}x$; rearranging isolates one of the two integrals in terms of the other.

Theorem 1.1 (Integration by parts). For continuously differentiable u and v,

$$\int u\,v'\,\mathrm{d}x = uv - \int u'\,v\,\mathrm{d}x,$$

and, for a definite integral,

$$\int_a^b u\,v'\,\mathrm{d}x = [uv]_a^b - \int_a^b u'\,v\,\mathrm{d}x.$$

The method trades the integral $\int uv'$ for the integral $\int u'v$, which pays off only if the new integral is *easier*—or, as in the cyclic example below, if it brings the original integral back and lets us solve for it. Success therefore hinges on the split: we differentiate u (so it should simplify) and integrate v' (so it should be no worse). A reliable order of preference for the part to call u is the mnemonic **LIATE**.

Note (Choosing u: **LIATE**). Pick u to be whichever factor comes first in the list

Logarithm $\to$ **I**nverse trig $\to$ **A**lgebraic (x^n) $\to$ **T**rigonometric $\to$ **E**xponential,

and let the rest be v'. Differentiating the earlier kinds simplifies them; integrating the later kinds is easy.

Example 1.1 ***(The basic case: $\int x\,\mathrm{e}^x\,\mathrm{d}x$).*** Here x (algebraic) precedes e^x (exponential), so $u = x$ and $v' = \mathrm{e}^x$, giving $u' = 1$ and $v = \mathrm{e}^x$:

$$\int x\,\mathrm{e}^x\,\mathrm{d}x = x\,\mathrm{e}^x - \int 1\cdot\mathrm{e}^x\,\mathrm{d}x = x\,\mathrm{e}^x - \mathrm{e}^x + C = (x-1)\,\mathrm{e}^x + C.$$

The new integral $\int \mathrm{e}^x\,\mathrm{d}x$ is immediate—the point of the method.

Example 1.2 ***(A function that is its own by-parts problem:*** $\int \ln x \; \mathrm{d}x$***).*** There is no obvious antiderivative of $\ln x$, but parts supplies one. Logarithm leads the list, so $u = \ln x$ and $v' = 1$, giving $u' = \frac{1}{x}$ and $v = x$:

$$\int \ln x \; \mathrm{d}x = x \ln x - \int x \cdot \frac{1}{x} \; \mathrm{d}x = x \ln x - \int 1 \; \mathrm{d}x = x \ln x - x + C.$$

The same trick—taking $v' = 1$—integrates $\arctan x$ and the other inverse functions.

Example 1.3 ***(Parts applied twice: a cyclic integral).*** For $I = \displaystyle\int \mathrm{e}^x \cos x \; \mathrm{d}x$, take $u = \cos x$, $v' = \mathrm{e}^x$:

$$I = \mathrm{e}^x \cos x + \int \mathrm{e}^x \sin x \; \mathrm{d}x.$$

Applying parts again to the new integral ($u = \sin x$, $v' = \mathrm{e}^x$) gives $\int \mathrm{e}^x \sin x \; \mathrm{d}x = \mathrm{e}^x \sin x - I$, so the original integral reappears:

$$I = \mathrm{e}^x \cos x + \mathrm{e}^x \sin x - I \quad \Longrightarrow \quad 2I = \mathrm{e}^x(\cos x + \sin x),$$

hence $I = \frac{1}{2}\mathrm{e}^x(\cos x + \sin x) + C$. When parts returns the integral you began with, solve for it algebraically rather than continuing. (Such integrals govern the response of a damped oscillator or an AC circuit.)

1.2 Partial fractions

A *rational function*—a ratio of polynomials—can be split into a sum of simpler fractions, each of which is on the standard-integral table or yields a logarithm or an arctangent. This is the reverse of putting fractions over a common denominator.

Note (The partial-fraction recipe). To integrate $\dfrac{P(x)}{Q(x)}$:

1. If $\deg P \geq \deg Q$ (an *improper* fraction), first divide to get a polynomial plus a proper remainder.
2. Factorise Q into linear and irreducible-quadratic factors, and write one partial fraction per factor:

$$\frac{A}{ax+b}, \qquad \frac{A}{ax+b} + \frac{B}{(ax+b)^2} \text{ (repeated)}, \qquad \frac{Ax+B}{ax^2+bx+c} \text{ (irreducible quadratic)}.$$

3. Find the unknown constants by clearing denominators and matching, then integrate each piece.

Example 1.4 ***(Two distinct linear factors).*** For $\displaystyle\int \frac{5x-4}{(x-2)(x+1)} \; \mathrm{d}x$, write

$$\frac{5x-4}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1} \quad \Longrightarrow \quad 5x - 4 = A(x+1) + B(x-2).$$

Setting $x = 2$ gives $6 = 3A$, so $A = 2$; setting $x = -1$ gives $-9 = -3B$, so $B = 3$. Hence

$$\int \frac{5x-4}{(x-2)(x+1)}\,\mathrm{d}x = \int\Big(\frac{2}{x-2} + \frac{3}{x+1}\Big)\,\mathrm{d}x = 2\ln|x-2| + 3\ln|x+1| + C.$$

Note. An irreducible quadratic factor $\frac{Ax+B}{ax^2+bx+c}$ leads to a logarithm, an arctan, or both: the part matching the derivative of the denominator integrates to a ln, and what is left gives an arctan after completing the square, using $\int \frac{\mathrm{d}x}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$ from Unit 9; a repeated factor $(ax+b)^2$ contributes a term integrating to $-\frac{1}{a(ax+b)}$. The recipe is the same in every case: break the fraction up, then integrate the pieces one at a time.

Example 1.5 ***(A repeated factor: cover-up is not enough).*** Evaluate $\int \frac{\mathrm{d}x}{x(x+1)^2}$. The repeated factor demands a term for *each* power up to the highest, so

$$\frac{1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} \quad\Longrightarrow\quad 1 = A(x+1)^2 + Bx(x+1) + Cx.$$

The cover-up trick still reaches the two "top" constants: setting $x = 0$ gives $A = 1$, and setting $x = -1$ gives $C = -1$. But cover-up cannot see B: its term $Bx(x+1)$ carries both factors x and $x+1$, so it vanishes at $x = 0$ *and* at $x = -1$ and neither substitution says anything about it. Match another coefficient instead: the x^2 terms give $0 = A + B$, hence $B = -1$ (equivalently, substitute any third value of x). Thus

$$\frac{1}{x(x+1)^2} = \frac{1}{x} - \frac{1}{x+1} - \frac{1}{(x+1)^2},$$

and integrating term by term—the last piece via the $-\frac{1}{a(ax+b)}$ rule of the note above—gives

$$\int \frac{\mathrm{d}x}{x(x+1)^2} = \ln|x| - \ln|x+1| + \frac{1}{x+1} + C = \ln\,|\,[\,|\,\Big]\frac{x}{x+1} + \frac{1}{x+1} + C.$$

Cover-up alone settles only the highest power of each repeated factor; the remaining constants need a coefficient match or a substituted value.

Example 1.6 ***(An irreducible quadratic: completing the square).*** The denominator of $\int \frac{\mathrm{d}x}{x^2+2x+5}$ does not factor over the reals. Complete the square, $x^2+2x+5 = (x+1)^2+4$, and substitute $u = x+1$:

$$\int \frac{\mathrm{d}x}{x^2+2x+5} = \int \frac{\mathrm{d}u}{u^2+4} = \tfrac{1}{2}\arctan\frac{u}{2} + C = \tfrac{1}{2}\arctan\frac{x+1}{2} + C.$$

Example 1.7 ***(An improper fraction: divide first).*** When the numerator's degree is at least the denominator's, divide before splitting. For $\int \frac{x^2}{x^2+1}\,\mathrm{d}x$, polynomial division gives

$\frac{x^2}{x^2+1} = 1 - \frac{1}{x^2+1}$, so

$$\int \frac{x^2}{x^2+1}\,\mathrm{d}x = \int \Big(1 - \frac{1}{x^2+1}\Big)\,\mathrm{d}x = x - \arctan x + C.$$

1.3 Trigonometric integrals

Powers and products of trigonometric functions yield to the Pythagorean identity of Unit 1 and the double-angle identities of Unit 3 before any integration is attempted. Two families cover most cases.

Powers of sine and cosine

An *even* power is reduced with the half-angle identities

$$\cos^2 x = \tfrac{1}{2}\big(1 + \cos 2x\big), \qquad \sin^2 x = \tfrac{1}{2}\big(1 - \cos 2x\big),$$

while an *odd* power is handled by splitting off one factor and converting the rest with $\sin^2 x + \cos^2 x = 1$, ready for a substitution.

Example 1.8 ***(An even and an odd power).*** Using the half-angle identity,

$$\int \cos^2 x\,\mathrm{d}x = \int \tfrac{1}{2}\big(1 + \cos 2x\big)\,\mathrm{d}x = \tfrac{1}{2}x + \tfrac{1}{4}\sin 2x + C.$$

For the odd power, split off one $\sin x$ and write the rest in cosines; with $u = \cos x,\ \mathrm{d}u = -\sin x\,\mathrm{d}x$,

$$\int \sin^3 x\,\mathrm{d}x = \int (1 - \cos^2 x)\sin x\,\mathrm{d}x = -\int (1 - u^2)\,\mathrm{d}u = -u + \tfrac{1}{3}u^3 + C,$$

which in terms of x is $-\cos x + \frac{1}{3}\cos^3 x + C$.

Powers of tangent and secant

These use the Pythagorean identity in the form $1 + \tan^2 x = \sec^2 x$, together with the standard results $\frac{\mathrm{d}}{\mathrm{d}x}\tan x = \sec^2 x$ and (from Unit 9) $\int \sec^2 x\,\mathrm{d}x = \tan x$.

Example 1.9 **(*$\int \tan^2 x\,\mathrm{d}x$ and $\int \tan^3 x\,\mathrm{d}x$*).** For the square, replace $\tan^2 x$ by $\sec^2 x - 1$:

$$\int \tan^2 x\,\mathrm{d}x = \int \big(\sec^2 x - 1\big)\,\mathrm{d}x = \tan x - x + C.$$

For an odd power, split off $\tan^2 x = \sec^2 x - 1$ and substitute $u = \sec x$ in the part that needs it (using $\int \tan x\,\mathrm{d}x = \ln|\sec x| + C$):

$$\int \tan^3 x\,\mathrm{d}x = \int \tan x\,(\sec^2 x - 1)\,\mathrm{d}x = \tfrac{1}{2}\tan^2 x - \ln|\sec x| + C.$$

The same split—peel off $\sec^2 x$ or $\tan^2 x$ and convert the rest with $1 + \tan^2 x = \sec^2 x$—handles $\int \tan^m x \sec^n x\,\mathrm{d}x$ whenever there is something to peel: n even, or m odd. The leftover cases need their own trick, the one worth remembering being

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C$$

(multiply and divide by $\sec x + \tan x$ and substitute $u = \sec x + \tan x$), which the trigonometric substitutions below call on.

1.4 Trigonometric substitution

A square root of a quadratic, $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$ or $\sqrt{x^2 - a^2}$, can be cleared by a substitution that turns it into a single trigonometric function, through the identities $1 - \sin^2\theta = \cos^2\theta$ and $1 + \tan^2\theta = \sec^2\theta$.

Note (The three trigonometric substitutions).

$$\sqrt{a^2 - x^2} \;\to\; x = a\sin\theta, \qquad \sqrt{a^2 + x^2} \;\to\; x = a\tan\theta, \qquad \sqrt{x^2 - a^2} \;\to\; x = a\sec\theta.$$

After integrating in θ, return to x with the original substitution and a right triangle.

Example 1.10 ***(Recovering an inverse-sine integral).*** For $\int \frac{dx}{\sqrt{4 - x^2}}$ (so $a = 2$), put $x = 2\sin\theta$ with $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, giving $dx = 2\cos\theta \, d\theta$ and $\sqrt{4 - x^2} = \sqrt{4 - 4\sin^2\theta} = 2|\cos\theta| = 2\cos\theta$ (the range makes $\cos\theta \geq 0$, and also makes $\theta = \arcsin\frac{x}{2}$ single-valued). The root cancels:

$$\int \frac{dx}{\sqrt{4 - x^2}} = \int \frac{2\cos\theta}{2\cos\theta}\, d\theta = \int d\theta = \theta + C = \arcsin\frac{x}{2} + C,$$

the entry promised by the standard-integral table of Unit 9.

Example 1.11 ***(The secant case, with a triangle to return).*** The previous root cancelled outright; the general mechanism instead leaves a trigonometric function of θ that must be carried back to x. Evaluate $\int \frac{dx}{x^2\sqrt{x^2 - a^2}}$ (with $x > a > 0$). The form $\sqrt{x^2 - a^2}$ calls for $x = a\sec\theta$, $\theta \in [0, \frac{\pi}{2})$, so that $dx = a\sec\theta\tan\theta \, d\theta$, $x^2 = a^2\sec^2\theta$ and

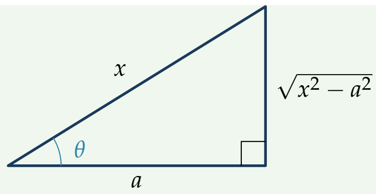
$$\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = a\sqrt{\sec^2\theta - 1} = a\tan\theta,$$

using $\sec^2\theta - 1 = \tan^2\theta$ (and $\tan\theta \geq 0$ on the chosen range). The integrand collapses neatly:

$$\int \frac{dx}{x^2\sqrt{x^2 - a^2}} = \int \frac{a\sec\theta\tan\theta}{a^2\sec^2\theta \cdot a\tan\theta}\, d\theta = \frac{1}{a^2}\int \frac{d\theta}{\sec\theta} = \frac{1}{a^2}\int \cos\theta \, d\theta = \frac{1}{a^2}\sin\theta + C,$$

needing only $\int \cos\theta \, d\theta$. To return to x, read $\sin\theta$ from a right triangle built from $\sec\theta = x/a$ (hypotenuse x, adjacent a, so the opposite side is $\sqrt{x^2 - a^2}$):

$$\sin\theta = \frac{\sqrt{x^2 - a^2}}{x} \quad\Longrightarrow\quad \int \frac{dx}{x^2\sqrt{x^2 - a^2}} = \frac{\sqrt{x^2 - a^2}}{a^2\, x} + C.$$



The triangle is the device the cancelling example never needed: substitute, simplify with a Pythagorean identity, integrate, then *convert back* through the triangle that encodes the original substitution.

Remark (The other two forms). The substitution $x = a\tan\theta$ turns $a^2 + x^2$ into $a^2\sec^2\theta$ and gives, for instance, $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$; the related $\int \frac{dx}{\sqrt{a^2+x^2}}$ evaluates to $\ln\left(x + \sqrt{a^2+x^2}\right) + C$ (also written $\operatorname{arsinh}\frac{x}{a}$, though we keep the logarithm form to the fore). The two worked examples above—the cancelling sine case and the secant case with its triangle—show the mechanism; every such integral follows the same three steps: substitute, simplify with a Pythagorean identity, and convert back.

2 Applications of the definite integral

The definite integral was built as an area; we now use it for areas, for the volumes that areas sweep out, for averages, and—reading an area as a probability—for the expected value of a random quantity.

2.1 Area between two curves

The signed area under one curve generalises at once to the area *between* two. If $f(x) \geq g(x)$ throughout $[a, b]$, a thin strip at x has height $f(x) - g(x)$ and width dx, so summing the strips gives the enclosed area.

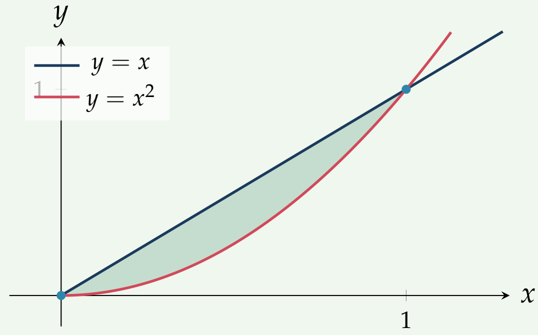
Theorem 2.1 (Area between curves). If f and g are continuous with $f(x) \geq g(x)$ on $[a, b]$, the area between the curves from $x = a$ to $x = b$ is

$$A = \int_a^b \left(f(x) - g(x)\right)\, dx \qquad (\text{top} - \text{bottom}).$$

Where the curves cross, split the interval and integrate each piece with the correct function on top.

Example 2.1 ***(Between a line and a parabola).*** The line $y = x$ and the parabola $y = x^2$ meet where $x = x^2$, i.e. at $x = 0$ and $x = 1$. Between them the line is on top ($x \geq x^2$ for $0 \leq x \leq 1$), so

$$A = \int_0^1 (x - x^2)\, dx = \left[\tfrac{1}{2}x^2 - \tfrac{1}{3}x^3\right]_0^1 = \tfrac{1}{2} - \tfrac{1}{3} = \frac{1}{6}.$$



2.2 Volumes of revolution

Spin the region under $y = f(x)$ about the x-axis and it sweeps out a solid. A thin strip of height $y = f(x)$ becomes a *disc* of radius y and thickness $\mathrm{d}x$, of volume $\pi y^2\,\mathrm{d}x$; summing the discs gives the volume.

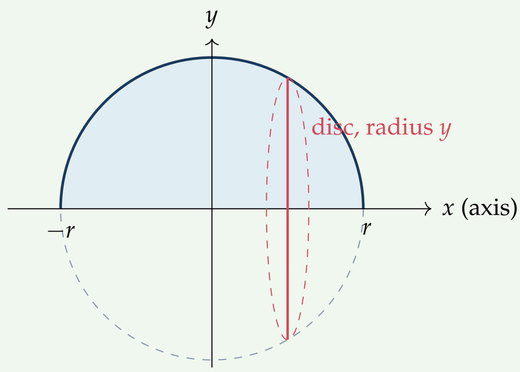
Theorem 2.2 (Disc method). The solid obtained by revolving the region under $y = f(x)$, $a \le x \le b$, about the x-axis has volume

$$V = \pi \int_a^b \big(f(x)\big)^2\,\mathrm{d}x.$$

Example 2.2 ***(The volume of a sphere).*** A sphere of radius r is the solid swept by the semicircular region under $y = \sqrt{r^2 - x^2}$, $-r \le x \le r$, revolved about the x-axis. Since $y^2 = r^2 - x^2$,

$$V = \pi \int_{-r}^{r} \big(r^2 - x^2\big)\,\mathrm{d}x = \pi\Big[r^2 x - \tfrac{1}{3}x^3\Big]_{-r}^{r} = \pi\Big(\tfrac{2r^3}{3} - \big(-\tfrac{2r^3}{3}\big)\Big) = \frac{4}{3}\pi r^3,$$

the familiar formula, now derived rather than quoted.



2.3 The average value of a function

The mean of finitely many numbers is their sum over their count. For a function on $[a, b]$, the "sum" is an integral and the "count" is the length $b - a$.

Definition 2.1 (Average value). The *average value* of a continuous f over $[a, b]$ is

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x)\, \mathrm{d}x.$$

Equivalently, $\bar{f}$ is the height of the rectangle on $[a, b]$ whose area equals the area under f.

Example 2.3 ***(A parabola's average).*** The average value of $f(x) = 4 - x^2$ over $[0, 2]$ is

$$\bar{f} = \frac{1}{2-0} \int_0^2 (4 - x^2)\, \mathrm{d}x = \frac{1}{2}\left[4x - \tfrac{1}{3}x^3\right]_0^2 = \frac{1}{2}\left(8 - \tfrac{8}{3}\right) = \frac{8}{3}.$$

The equal-area rectangle's top is drawn meeting the graph, but does f actually *attain* the height $\bar{f}$ somewhere? For a continuous function it does, and this is worth recording.

Theorem 2.3 (Mean value theorem for integrals). If f is continuous on $[a, b]$, there is a point $c \in [a, b]$ with $f(c) = \bar{f}$; that is,

$$\int_a^b f(x)\, \mathrm{d}x = f(c)\,(b-a).$$

Proof. Being continuous on a closed interval, f attains a minimum m and a maximum M there (Unit 8). Since $m \le f(x) \le M$ on all of $[a, b]$, integrating preserves the inequalities, $m(b-a) \le \int_a^b f \le M(b-a)$, and dividing by $b - a > 0$ gives $m \le \bar{f} \le M$. The intermediate value theorem (Unit 8) says a continuous f takes every value between its minimum and maximum; in particular it takes the value $\bar{f}$ at some $c \in [a, b]$. ∎

This is precisely why the equal-area rectangle's top really does cross the graph: continuity forces the average to be a genuine height of f, not merely a number between its extremes.

The average value weights every point of $[a, b]$ equally. If instead some outcomes are more likely than others, we weight by a *density*—and the average becomes an *expectation*. To express it for quantities with no upper bound, we first need integrals over an infinite range.

2.4 Improper integrals, probability and expectation

Definition 2.2 (Improper integral). An integral over an unbounded interval is defined as a limit of ordinary integrals,

$$\int_a^\infty f(x)\, \mathrm{d}x = \lim_{t\to\infty} \int_a^t f(x)\, \mathrm{d}x, \qquad \int_{-\infty}^b f(x)\, \mathrm{d}x = \lim_{s\to-\infty} \int_s^b f(x)\, \mathrm{d}x.$$

If the limit is finite the integral *converges*; otherwise it *diverges*. A doubly infinite integral is *split* at any point c,

$$\int_{-\infty}^\infty f(x)\, \mathrm{d}x = \int_{-\infty}^c f(x)\, \mathrm{d}x + \int_c^\infty f(x)\, \mathrm{d}x,$$

and converges only when *each* piece converges on its own—not as the single symmetric limit $\lim_{t\to\infty} \int_{-t}^{t} f$, which can be finite even when the two halves separately diverge.

Example 2.4 ***(Convergence and divergence).***

$$\int_1^\infty \frac{\mathrm{d}x}{x^2} = \lim_{t\to\infty}\left[-\frac{1}{x}\right]_1^t = \lim_{t\to\infty}\left(1-\frac{1}{t}\right) = 1,$$

a finite area under an infinitely long tail. By contrast $\int_1^\infty \frac{\mathrm{d}x}{x} = \lim_{t\to\infty}[\ln x]_1^t = \lim_{t\to\infty} \ln t = \infty$ diverges: the tail of $1/x$ is just too fat. Whether the area under a tail is finite is a genuine question, and these integrals are met throughout physics and data science.

Probability densities and expectation

A continuous random quantity X is described by a *probability density* f, with $f(x) \geq 0$ and total area one, $\int_{-\infty}^{\infty} f = 1$. The probability that X lands in an interval is the area over it, $P(a \leq X \leq b) = \int_a^b f(x)\,\mathrm{d}x$, and the *expectation* (mean) is the density-weighted average

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x\,f(x)\,\mathrm{d}x,$$

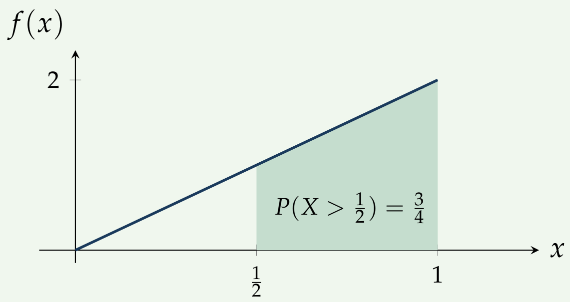
taken over the whole line and assumed to converge. This is the average value of the previous section with each x weighted by how likely it is.

Example 2.5 ***(A density on a bounded interval).*** Let $f(x) = 2x$ on $[0,1]$ (and 0 elsewhere). It is a valid density, since $f \geq 0$ and $\int_0^1 2x\,\mathrm{d}x = \left[x^2\right]_0^1 = 1$. The probability that X exceeds $\frac{1}{2}$ is the shaded area

$$P(X > \tfrac{1}{2}) = \int_{1/2}^1 2x\,\mathrm{d}x = \left[x^2\right]_{1/2}^1 = 1 - \tfrac{1}{4} = \tfrac{3}{4},$$

and the expectation is

$$\mathbb{E}[X] = \int_0^1 x \cdot 2x\,\mathrm{d}x = \int_0^1 2x^2\,\mathrm{d}x = \left[\tfrac{2}{3}x^3\right]_0^1 = \tfrac{2}{3}.$$



Example 2.6 ***(A density on an infinite range: the exponential law).*** Waiting times and radioactive lifetimes follow the *exponential* density $f(x) = \lambda \mathrm{e}^{-\lambda x}$ on $[0, \infty)$, with rate $\lambda > 0$. It integrates to one,

$$\int_0^\infty \lambda \mathrm{e}^{-\lambda x}\,\mathrm{d}x = \lim_{t\to\infty}\left[-\mathrm{e}^{-\lambda x}\right]_0^t = 0 - (-1) = 1,$$

and its mean follows by parts ($u = x$, $v' = \lambda e^{-\lambda x}$, so $v = -e^{-\lambda x}$):

$$\mathbb{E}[X] = \int_0^\infty x\,\lambda e^{-\lambda x}\,dx = \left[-x\,e^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\,dx = 0 + \frac{1}{\lambda} = \frac{1}{\lambda}.$$

The mean lifetime is the reciprocal of the rate—the by-parts technique of Section 1 and the improper integral working together.

Remark (Back to the Gaussian). The most important density of all, the normal (bell) curve built from e^{-x^2}, has *no* elementary antiderivative—the Liouville note of Unit 9. Its probabilities therefore cannot *in general* be written in closed form and are computed numerically, through the error function erf (symmetry still settles special cases exactly—half the area lies either side of the centre). The bell curve closes a thread begun with the sigmoid in Unit 1.

3 Setting up integrals from a description

The hardest part of an applied integral is rarely the integration; it is the *setting up*—turning a geometric or physical description into $\int_a^b (\cdots)\,dx$. Every example above follows one pattern, worth stating on its own.

Note (A recipe for setting up an integral).

1. Slice the quantity into thin pieces indexed by a variable (here x), each of "width" dx.
2. Write the contribution of one typical slice as (something) $\times$ dx, where the "something" is evaluated at x—a height, an area, a density, a force.
3. Add the slices by integrating over the range the variable covers.

Example 3.1 ***(Mass of a rod of variable density).*** A rod occupies $0 \le x \le L$ with linear density $\rho(x) = \rho_0(1 + \frac{x}{L})$ (it is denser towards one end). A slice of length dx at position x has mass $dm = \rho(x)\,dx$, so the total mass is

$$M = \int_0^L \rho_0\left(1 + \frac{x}{L}\right)dx = \rho_0\left[x + \frac{x^2}{2L}\right]_0^L = \rho_0\left(L + \frac{L}{2}\right) = \frac{3}{2}\rho_0 L.$$

The same three steps—slice, weight, integrate—produced the area, volume and probability integrals above. They work just as well when the "something" carried by each slice is itself a varying *area*, or a *distance*.

Example 3.2 ***(Volume by cross-sections: a pyramid).*** The disc method is one case of a wider idea: if a solid has known cross-sectional area $A(x)$ perpendicular to the x-axis, a slab of thickness dx has volume $A(x)\,dx$, so $V = \int A(x)\,dx$. Take a pyramid of height h on a square base of side a, and measure x *down from the apex*. By similar triangles the square cross-section at depth x has side $a\,\frac{x}{h}$, and hence area $A(x) = a^2\,\frac{x^2}{h^2}$, so

$$V = \int_0^h a^2\,\frac{x^2}{h^2}\,dx = \frac{a^2}{h^2}\left[\frac{x^3}{3}\right]_0^h = \frac{a^2 h}{3} = \tfrac{1}{3}\,(\text{base area}) \times (\text{height}),$$

the classical pyramid formula—and here the cross-section is a square, a shape no revolution could produce.

Example 3.3 ***(Work done lifting a hanging rope).*** A uniform rope of length L and mass μ per unit length hangs from the top of a building; how much work is needed to wind it all to the top? The piece of rope a distance x below the top has mass $\mu\ \mathrm{d}x$ and must be raised a distance x against gravity g, so it contributes work $\mathrm{d}W = (\mu\ \mathrm{d}x)\, g\, x$. Summing the pieces,

$$W = \int_0^L \mu g\, x\ \mathrm{d}x = \mu g \left[\frac{x^2}{2}\right]_0^L = \tfrac{1}{2}\,\mu g L^2.$$

The answer is the rope's total weight $\mu g L$ times the height $\frac{L}{2}$ of its midpoint—as if the whole weight hung from its *centre of mass*. That centre of mass, $\int x\ \mathrm{d}m / \int\ \mathrm{d}m$, is itself a density-weighted average of position—the same construction as the expectation $\mathbb{E}[X]$ of Section 2.

Remark. The recipe is indifferent to the discipline: a slice carries mass for the physicist, probability for the statistician, area or volume for the geometer. Recognising that shared structure—one quantity assembled by summing infinitely many infinitesimal contributions—is the real content of integral calculus, and the reason it reaches so far beyond the area problem that introduced it in Unit 9.

Summary

Concept	Key idea
By parts	$\int u\,v' = uv - \int u'\,v$ (product rule reversed). Choose u by **LIATE**; if the integral reappears, solve for it.
Partial fractions	Split a proper rational function into $\frac{A}{ax+b}, \frac{B}{(ax+b)^2}, \frac{Ax+B}{ax^2+bx+c}$ pieces; divide first if improper. Linear $\to$ ln, quadratic $\to$ ln and/or arctan.
Trig powers	Even powers via half-angle $\cos^2 = \frac{1}{2}(1+\cos 2x)$, $\sin^2 = \frac{1}{2}(1-\cos 2x)$; odd powers by splitting off one factor and using $\sin^2+\cos^2 = 1$; tangents/secants via $1+\tan^2 = \sec^2$.
Trig substitution	$\sqrt{a^2-x^2} \to a\sin\theta$, $\sqrt{a^2+x^2} \to a\tan\theta$, $\sqrt{x^2-a^2} \to a\sec\theta$; simplify with a Pythagorean identity, then convert back.
Area between curves	$A = \int_a^b (\text{top} - \text{bottom})\ \mathrm{d}x$; split where the curves cross.
Volume of revolution	Disc method: $V = \pi \int_a^b \big(f(x)\big)^2\,\mathrm{d}x$ about the x-axis.
Average value	$\bar{f} = \dfrac{1}{b-a}\int_a^b f$: the equal-area rectangle height.
Improper integral	$\int_a^\infty f = \lim_{t\to\infty}\int_a^t f$; converges if finite ($\int_1^\infty x^{-2} = 1$), else diverges ($\int_1^\infty x^{-1}$).
Density & expectation	$f \geq 0$, $\int f = 1$; $P(a \leq X \leq b) = \int_a^b f$ (an area); $\mathbb{E}[X] = \int x\,f(x)\ \mathrm{d}x$ (a density-weighted average).
Setting up	Slice into pieces (something) $\times$ dx, then integrate over the range.

Looking back. This completes the module. We began with the language of functions and the number systems they inhabit, crossed into the complex plane, then built the linear algebra *of vectors, matrices and eigensystems, and closed with the* differential and integral calculus *of one variable. Each strand has a sequel: the linear algebra continues into a full course where the eigenvalues return as the spectral theorem and the singular value decomposition; the calculus extends to several variables, series and differential equations in the dedicated Calculus module; and for physicists the same tools reappear as normal modes, work integrals and equations of motion. The toolkit assembled across these ten units is the shared foundation beneath all of it.*

Companion material: a separate *Exercise Sheet 10* (with solutions) develops the practice problems for this unit.